Phase changes

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for energy transfer during a phase change without a temperature rise.

SUCCESS CRITERIA - I can...

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Two objects at 280 K have different masses and materials. Must their internal energies be equal? Give two reasons.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: Q = mL; Q = mcΔT.
Use mL during a phase change at constant temperature, and mcΔT outside that interval. L is latent heat.

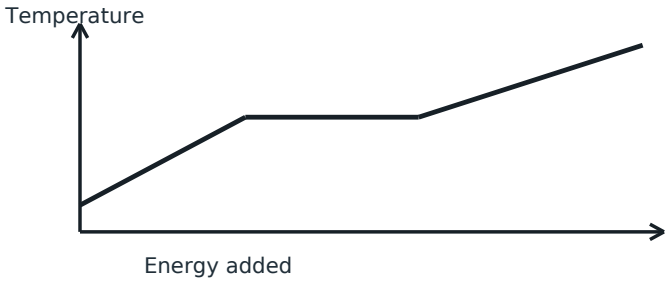
Worked reasoning: Melting 2 kg of a substance with latent heat 100 J/kg requires 200 J. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

Your turn: Sketch temperature versus added energy through warming and melting. Mark the plateau.

Marzano check - circle: ① ② ③

03 Represent and interpret

On the heating curve, identify the flat interval. Explain why added energy does not raise temperature during that interval.



The horizontal axis shows energy added. The vertical axis shows temperature.

Marzano check - circle: ① ② ③

04 Solve and justify

Melt .10 kg of ice already at 0°C. Use L_f=334, 000 J/kg. Then warm the resulting water to 20°C with c=4200 J/(kg·°C). Find each energy and total.

Marzano check - circle: ① ② ③

05 Analyze the reasoning

A student uses $mc\Delta T$ for a melting interval with $\Delta T=0$ and concludes no energy is required. Explain the missing term.

Marzano check - circle: 1 2 3

06 Choose a method

With right positive, a spring has $k=200$ N/m and displacement $x=-0.06$ m. Find signed spring force and direction.

Marzano check - circle: 1 2 3

07 Exit check

A .20 kg liquid releases 45, 000 J while condensing at constant temperature. Find L and state the sign of Q for the sample.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 0.2 kg ice sample at 0°C melts and the water warms to 10°C. Use $L_f=334000$ J/kg and $c_{water}=4200$ J/(kg·°C). Find total heat.

Marzano check - circle: 1 2 3